

AeroBook: AI Digital Textbook & Study Platform

Comprehensive Features & Portals Documentation

AeroBook is a modern, AI-powered interactive digital textbook and study platform. It personalizes and gamifies the learning experience for students, while providing detailed progress insights and complete curriculum control to teachers.

Below is a complete, user-friendly breakdown of the technology used, the system workflow, and the features of both the Student Portal and Teacher Portal.

High-Level Tech Stack (Technologies Used)

AeroBook uses modern technologies to deliver a fast, dynamic, and secure experience:

1. Frontend (User Interface)

- **React (Vite):** Powering a fast, single-page application with smooth transitions and fast page loading.
- **Tailwind CSS:** Used for building a premium, clean, and responsive user interface.
- **Zustand:** A lightweight state management library to manage global client state (user authentication, preferences, etc.).

2. Backend (Server & API Logic)

- **Node.js & Express:** Handles server-side routing, request validation, and database operations efficiently.
- **MongoDB (Mongoose):** Flexible database to store curriculum chapters, concepts, student records, and attempt logs.
- **Redis & Bull Queue:** Used for caching database queries and running asynchronous background tasks (generating reports, sending emails).
- **Socket.io (WebSockets):** Enables real-time communications, such as streaming AI tutor responses word-by-word.

3. AI Core (Artificial Intelligence)

- **Gemini & OpenAI API:** Drives the core AI features — the Personal AI Tutor, automated question generation, notes summarization, and monthly reports.
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Student Portal (Student Features)

The Student Portal is designed to keep learning engaging, structured, and customized.

Interactive Textbook & Learn Page

- **What it does:** Organizes the curriculum into a clean hierarchy: **Courses → Chapters → Concepts**. Each concept is an atomic unit of learning.
- **Benefits:** Students can read textual content and view associated rich media files (videos, animations, images) for better understanding.

Aeronaut (Personal AI Tutor)

- **What it does:** Students can chat with an AI tutor while reading any concept. It answers queries strictly using context from the textbook (RAG approach).
- **AI Explanation Modes:** Explains concepts in four formats:
 - **Visual:** Describes steps in visual/layout terms.
 - **Story:** Explains through an engaging narrative story.
 - **Steps:** Provides a numbered, simple walkthrough.
 - **Analogy:** Relates abstract concepts to real-world objects or scenarios.
- **Deep Thinking Modes:**
 - **Devil's Advocate:** Challenges the student's reasoning to develop critical thinking.
 - **Bridge:** Connects the concept to other subjects (e.g., linking Mathematics to Physics).

Smart Notes & Flashcards

- **What it does:** Automatically saves highlighted text from the textbook as digital notes.
- **Benefits:** AI summarizes these notes and automatically generates Q&A Flashcards for quick and active revision.

Adaptive Practice & Exercises

- **What it does:** Generates practice questions dynamically based on the student's mastery level.
- **Benefits:** Struggling students receive easier questions, and as they improve, the system automatically challenges them with harder questions. Includes hints and step-by-step solving feedback.

Gamification & Streaks

- **What it does:** Engages students through digital achievements and rewards.
- **Benefits:** Awarding XP (Experience Points) and badges (e.g., "7-Day Streak Master") for maintaining study streaks and completing chapters.

Dashboard & Heatmap

- **What it does:** Visualizes student progress.
- **Benefits:** Displays a GitHub-style green activity calendar showing daily study time. Includes a "Weak Area Identifier" that highlights topics that need more practice.

Interactive Polls & Quizzes (Practice Sets)

- **What it does:** Shows class tests or quizzes assigned by the teacher.
- **Benefits:** Students get instant results showing scored percentage (e.g., 🎉 85%), correct answers (✅), and incorrect answers (❌) upon submission.

Homework & Assignment Portal

- **What it does:** Allows students to view assigned homework, download question PDFs, and upload their completed answer sheets.
 - **Resubmission Buffer:** Allows students to re-upload files before grading if they make a mistake.
 - **Grading Feedback:** Displays grades, marks, and written feedback from the teacher directly on the assignment card.
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2 Teacher Portal (Teacher Features)

The Teacher Portal simplifies administration and gives teachers deep insights into classroom performance.

Teacher Dashboard

- **What it does:** Displays an aggregated class overview.
- **Benefits:** Shows active student count, total courses, class average accuracy, and pending assignment submissions at a glance.

Course & Chapter Creator

- **What it does:** Allows teachers to manage courses and build chapters.
- **Benefits:** Teachers can structure chapters, add concepts, attach media assets, define prerequisites, and set target difficulties.

Student Management & Directory

- **What it does:** Provides a detailed performance directory of the class.
- **Benefits:** Teachers can inspect individual student mastery levels, study streaks, active time, and exact concepts where they are struggling.

Assignment Manager (Upload & Grading)

- **What it does:** A hub for posting homework and grading submissions.
- **Benefits:** Teachers upload assignment PDFs, set due dates and max marks, and review submissions to assign scores and write feedback.

Custom Quiz & Poll Builder

- **What it does:** Creates custom MCQ practice tests.
- **Benefits:** Teachers can create questions manually or use the AI Generator to instantly draft relevant quizzes for any chapter.

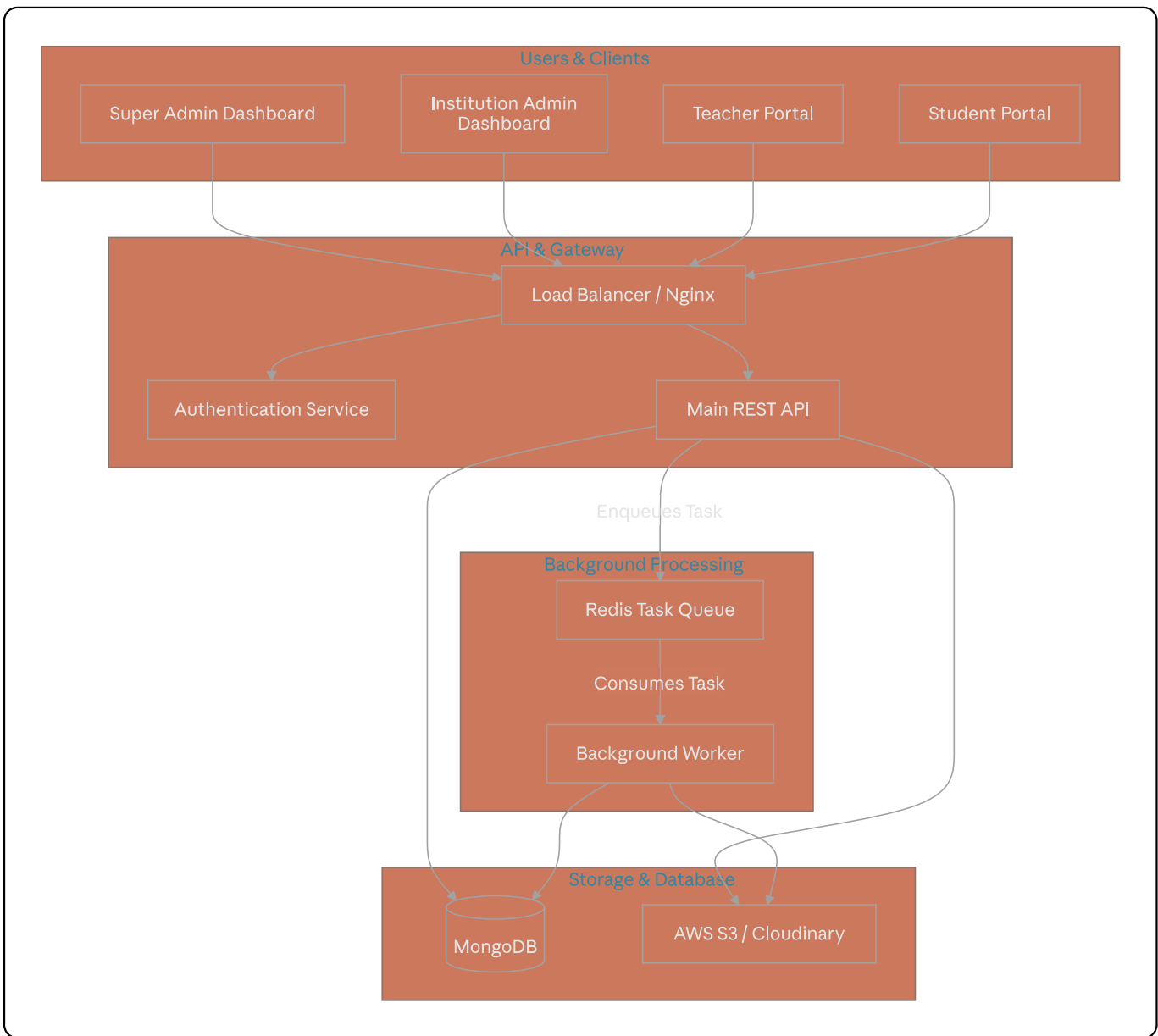
Practice Analytics

- **What it does:** Provides detailed analytics on quizzes and tests.
- **Benefits:** Visualizes class completion rates, average scores, and common areas where students struggled.

3. High-Level System Architecture & Flow

System Workflows

The diagram below shows how different portal users interact with the backend API services and database configurations:



4. Access Control & Role Matrix

This matrix shows the permissions of different roles across various features in the system:

Feature	Super Admin	Institution Admin	Teacher	Student
Manage Schools/Institutions	✓	✗	✗	✗
Global System Analytics	✓	✗	✗	✗
Manage Classes & Courses	✗	✓	✗	✗
Manage School Users (Single/Bulk)	✗	✓	✗	✗
View School Analytics	✗	✓	✗	✗
Manage & Create Chapters/Concepts	✗	✗	✓	✗
Create and Assign Custom Quizzes/Polls	✗	✗	✓	✗
Upload Homework & Assignments	✗	✗	✓	✗
Grade Student Assignments	✗	✗	✓	✗
View Student Progress Analytics	✗	✗	✓	✗
View Own Dashboard, Heatmaps & Grades	✗	✗	✗	✓
Submit Homework & Assignments	✗	✗	✗	✓
Attempt Quizzes & Use AI Aeronaut Tutor	✗	✗	✗	✓

5. Common / System Features

- **Auth & Profile Management:** Secure login using JWT authentication and customizable profile preferences.
- **Offline Sync Support:** Saves student progress and logs locally if the internet connection is lost. The data automatically syncs with the server when the user goes back online.